

FDM Solⁿ of an Eigen-Value Problem:



Problem: A slender column under an axial compressive load

Bending moment,

$$M = -Py$$

$$\therefore EI \frac{d^2 y}{dx^2} + Py = 0$$

This is the DEQ that governs the buckling shape of the column.

$$\frac{d^2 y}{dx^2} + \left(\frac{P}{EI}\right)y = 0$$

$$\Rightarrow \frac{d^2 y}{dx^2} + \lambda^2 y = 0 \quad \text{where } \lambda = \sqrt{\frac{P}{EI}}$$

The complementary function to the DEQ, $\frac{d^2 y}{dx^2} + \lambda^2 y = 0$; is

$$y(x) = A \cos \lambda x + B \sin \lambda x \quad \text{--- (ii)}$$

Known physical conditions:

$$\text{at } x=0, y=0 \quad \text{--- (a)}$$

$$\text{at } x=L, y=0 \quad \text{--- (b)}$$

$$\text{From (a), } A \cos 0 + B \sin 0 = 0$$

$$\Rightarrow A = 0$$

$$\therefore y(x) = B \sin \lambda x \quad \text{--- (iii)}$$

$$\text{From (b), } B \sin \lambda L = 0$$

Since, $B \neq 0$, $\sin \lambda L = 0$

$$0\pi, \sin \lambda L = \sin n\pi$$

$$\therefore \lambda L = n\pi$$

$$\Rightarrow \lambda = \frac{n\pi}{L}$$

$$0\pi \sqrt{\frac{P}{EI}} = \frac{n\pi}{L}$$

$$\therefore P = \frac{n^2 \pi^2 EI}{L^2} \text{ where } n = 1, 2, 3, \dots$$

$$1^{\text{st}} \text{ critical load, } P_{cr}^1 = \frac{\pi^2 EI}{L^2} \quad (n=1)$$

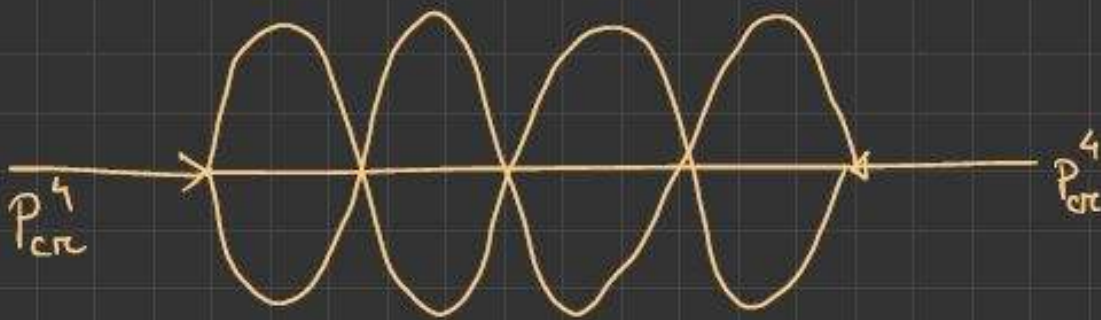
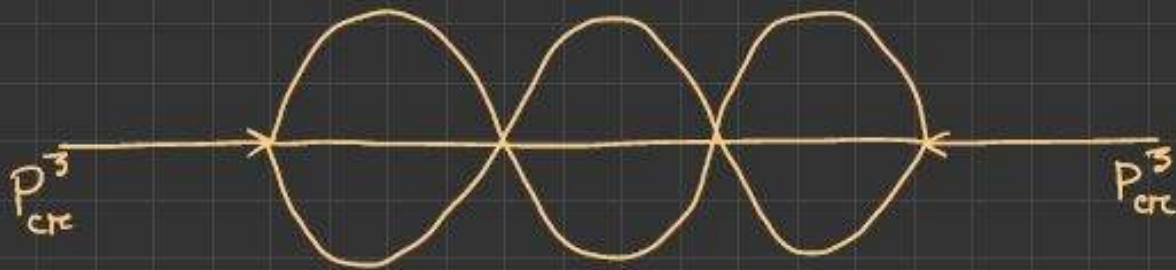
$$2^{\text{nd}} \text{ critical load, } P_{cr}^2 = \frac{4\pi^2 EI}{L^2} \quad (n=2)$$

$$3^{\text{rd}} \quad \quad \quad \quad \quad P_{cr}^3 = \frac{9\pi^2 EI}{L^2} \quad (n=3)$$

$$P_{cr}^4 = \frac{16\pi^2 EI}{L^2} \quad (n=4)$$

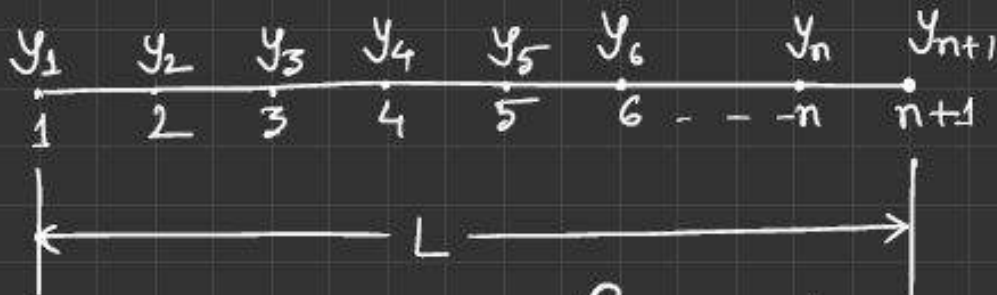
$\vdots$
 $\vdots$
 $\vdots$
 $\vdots$

Buckling loads



Buckling mode shape of column for different buckling loads.

FDM Solution:



$$\Delta x = h = \frac{L}{n}$$

$$x_1, x_2, x_3, \dots, x_{n+1}$$

n = no. of divisions

$$\begin{cases} y_1 = 0 \\ y_{n+1} = 0 \end{cases}$$

Given Boundary conditions

Governing eqn:

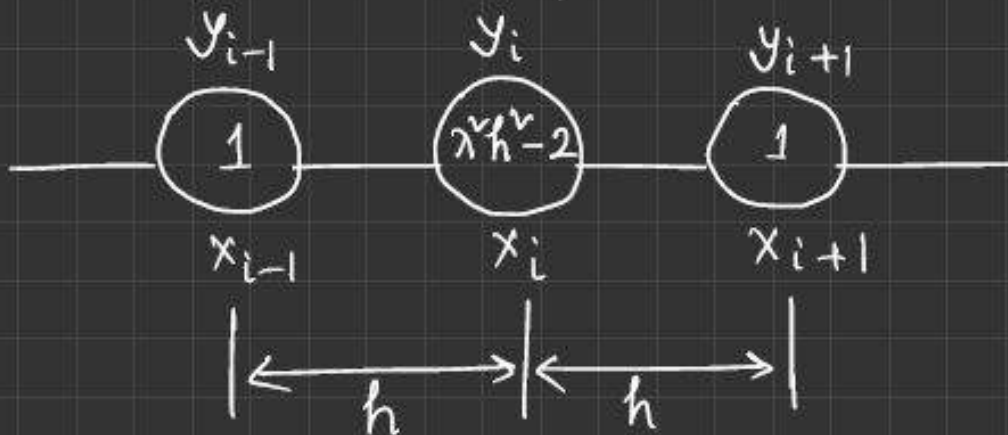
$$\frac{d^2 y}{dx^2} + \lambda^2 y = 0 \quad \text{--- (i)}$$

$$\Rightarrow \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2} + \lambda^2 y_i = 0$$

$$\Rightarrow y_{i-1} - 2y_i + y_{i+1} + \lambda^2 h^2 y_i = 0$$

$$\Rightarrow y_{i-1} + (\lambda^2 h^2 - 2)y_i + y_{i+1} = 0$$

--- (ii)



$$y_1 = 0 \quad \text{--- (1)}$$

$$y_1 + (\lambda^v h^v - 2)y_2 + y_3 = 0 \quad \text{--- (2)}$$

$$y_2 + (\lambda^v h^v - 2)y_3 + y_4 = 0 \quad \text{--- (3)}$$

$$y_3 + (\lambda^v h^v - 2)y_4 + y_5 = 0 \quad \text{--- (4)}$$

$$\vdots$$

$$y_{n-1} + (\lambda^v h^v - 2)y_n + y_{n+1} = 0 \quad \text{--- (n)}$$

$$y_{n+1} = 0 \quad \text{--- (n+1)}$$

A set of linear, homogeneous, algebraic solutions.
Unique solⁿ does not exist.

Case - I:
$$\begin{cases} y_1 = 0 \\ y_1 + (\lambda^v h^v - 2)y_2 + y_3 = 0 \\ y_3 = 0 \end{cases}$$

$\Rightarrow (\lambda^v h^v - 2)y_2 = 0$ Minimum no. of divisions that can be applied is 2.

Since, $y_2 \neq 0$; $\lambda^v h^v - 2 = 0$
 $\Rightarrow \lambda^v = \frac{2}{h^v}$

$$\therefore \frac{P}{EI} = \frac{2}{h^2}$$

$$\begin{aligned} \Rightarrow P_{cr} &= \frac{2EI}{h^2} \\ &= \frac{2EI}{\left(\frac{L}{2}\right)^2} \\ &= \frac{8EI}{L^2} \quad (\text{FDM}) \end{aligned}$$

$$P_{cr}^{\text{exact}} = \frac{9.87 EI}{L^2}$$

Case - II:

$$\left. \begin{aligned} y_1 &= 0 \\ y_1 + (\lambda^2 h^2 - 2)y_2 + y_3 &= 0 \\ y_2 + (\lambda^2 h^2 - 2)y_3 + y_4 &= 0 \\ y_4 &= 0 \end{aligned} \right\} n=3$$

Now, $(\lambda^2 h^2 - 2)y_2 + y_3 = 0$
 $y_2 + (\lambda^2 h^2 - 2)y_3 = 0$

$$\begin{bmatrix} \lambda^2 h^2 - 2 & 1 \\ 1 & \lambda^2 h^2 - 2 \end{bmatrix}$$

$$\Rightarrow \begin{cases} \left(\frac{2}{h^2} - \lambda^v\right) y_2 - \frac{1}{h^2} y_3 = 0 \\ -\frac{1}{h^2} y_2 + \left(\frac{2}{h^2} - \lambda^v\right) y_3 = 0 \end{cases}$$

$$\Rightarrow \begin{bmatrix} \frac{2}{h^2} - \lambda^v & -\frac{1}{h^2} \\ -\frac{1}{h^2} & \frac{2}{h^2} - \lambda^v \end{bmatrix} \begin{Bmatrix} y_2 \\ y_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\Rightarrow ([A] - \lambda^v [I]) \{y\} = \{0\}$$

$$\text{where } [A] = \begin{bmatrix} \frac{2}{h^2} & -\frac{1}{h^2} \\ -\frac{1}{h^2} & \frac{2}{h^2} \end{bmatrix}$$

For eigen values, $|[A] - \lambda^v [I]| = 0$

$$\begin{vmatrix} \frac{2}{h^2} - \lambda^v & -\frac{1}{h^2} \\ -\frac{1}{h^2} & \frac{2}{h^2} - \lambda^v \end{vmatrix} = 0$$

$$\Rightarrow \left(\frac{2}{h^2} - \lambda^2 \right)^2 - \frac{1}{h^4} = 0$$

$$\Rightarrow \frac{4}{h^4} - \frac{4}{h^2} \lambda^2 + \lambda^4 - \frac{1}{h^4} = 0$$

$$\Rightarrow \lambda^4 - \frac{4}{h^2} \lambda^2 + \frac{3}{h^4} = 0$$

$$\therefore \lambda^2 = \frac{\frac{4}{h^2} \pm \sqrt{\frac{16}{h^4} - \frac{12}{h^4}}}{2}$$

$$= \frac{2}{h^2} \pm \frac{1}{h^2}$$

$$= \frac{2 \pm 1}{h^2} = \frac{3}{h^2}, \frac{1}{h^2}$$

$$\lambda_1^2 = \frac{1}{h^2}$$

$$\frac{P_1}{EI} = \frac{9}{L^2}$$

$$P_1 = \frac{9EI}{L^2}$$

$$\text{And, } \lambda_2^2 = \frac{3}{h^2}$$

$$\frac{P_2}{EI} = \frac{27}{L^2}$$

$$\Rightarrow P_2 = \frac{27EI}{L^2}$$

For $n=4$:

$$\left. \begin{aligned} (\lambda^r h^r - 2)y_2 + y_3 &= 0 \\ y_2 + (\lambda^r h^r - 2)y_3 + y_4 &= 0 \\ y_3 + (\lambda^r h^r - 2)y_4 &= 0 \end{aligned} \right\}$$

$$\begin{bmatrix} \lambda^r h^r - 2 & 1 & 0 \\ 1 & \lambda^r h^r - 2 & 1 \\ 0 & 1 & \lambda^r h^r - 2 \end{bmatrix} \begin{Bmatrix} y_2 \\ y_3 \\ y_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2/h^r - \lambda^r & -1/h^r & 0 \\ -1/h^r & 2/h^r - \lambda^r & -1/h^r \\ 0 & -1/h^r & 2/h^r - \lambda^r \end{bmatrix} \begin{Bmatrix} y_2 \\ y_3 \\ y_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

$$\Rightarrow ([A] - \lambda[I])\{y\} = \{0\}$$

Now for eigen values,

$$\begin{vmatrix} 2/h^2 - \lambda^2 & -1/h^2 & 0 \\ -1/h^2 & 2/h^2 - \lambda^2 & -1/h^2 \\ 0 & -1/h^2 & 2/h^2 - \lambda^2 \end{vmatrix} = 0$$

$$\Rightarrow \left(\frac{2}{h^2} - \lambda^2 \right) \left[\left(\frac{2}{h^2} - \lambda^2 \right)^2 - \frac{1}{h^4} \right] + \frac{1}{h^2} \left[-\frac{1}{h^2} \left(\frac{2}{h^2} - \lambda^2 \right) \right] = 0$$

$$\Rightarrow \left(\frac{2}{h^2} - \lambda^2 \right) \left[\frac{4}{h^4} - \frac{4}{h^2} \lambda^2 + \lambda^4 - \frac{1}{h^4} - \frac{1}{h^4} \right] = 0$$

$$\Rightarrow \boxed{\lambda^2 = 2/h^2}$$

$$\text{And, } \lambda^4 - \frac{4}{h^2} \lambda^2 + \frac{2}{h^4} = 0$$

$$\Rightarrow \lambda^2 = \frac{4/h^2 \pm \sqrt{16/h^4 - 8/h^4}}{2}$$

$$\boxed{\lambda^2 = \frac{2 \pm \sqrt{2}}{h^2}}$$

$$\therefore P_1 = \frac{(2-\sqrt{2})EI}{h^2} = \frac{(2-\sqrt{2}) \times 16}{L^2} EI$$

$$= \frac{9.3726 EI}{L^2}$$

$$P_2 = \frac{2EI}{h^2} = \frac{32EI}{L^2}$$

$$P_3 = \frac{(2+\sqrt{2})EI}{h^2} = \frac{(2+\sqrt{2}) \times 16}{L^2} \times EI$$

$$= \frac{54.6274 EI}{L^2}$$

$n=3:-$

$$J_1 = 0, \quad J_1 + (\lambda^2 h^2 - 2)J_2 + J_3 = 0, \quad J_3 = 0$$

$$\therefore (\lambda^2 h^2 - 2)J_2 = 0$$

$$\Rightarrow E J_2 = 0 \quad [E = \lambda^2 h^2 - 2]$$

$$\therefore E = 0 \Rightarrow E_3 = 0$$

$$\therefore E_3 = E$$

$n=4:-$

$$J_1 = 0, \quad J_1 + (\lambda^2 h^2 - 2)J_2 + J_3 = 0, \quad J_2 + (\lambda^2 h^2 - 2)J_3 + J_4 = 0,$$

$$(\lambda^2 h^2 - 2)J_2 + J_3 = 0$$

$$J_2 + (\lambda^2 h^2 - 2)J_3 = 0 \Rightarrow J_2 = -(\lambda^2 h^2 - 2)J_3$$

$$-(\lambda^2 h^2 - 2)^2 J_3 + J_3 = 0$$

$$\Rightarrow (-E^2 + 1)J_3 = 0$$

$$\Rightarrow J_3 E_4 = 0 \quad \therefore E_4 = -E^2 + 1 = -E E_3 - E_2 \quad [E_2 = -1]$$

$n=5:-$

$$J_1 = 0, \quad J_1 + J_2(\lambda^2 h^2 - 2) + J_3 = 0$$

$$J_2 + (\lambda^2 h^2 - 2)J_3 + J_4 = 0$$

$$J_3 + (\lambda^2 h^2 - 2)J_4 + J_5 = 0$$

$$J_5 = 0$$

solving these Eq^{ns}:-

$$(\lambda^2 h^2 - 2)^3 J_4 - 2(\lambda^2 h^2 - 2) J_4 = 0$$

$$\Rightarrow (E^3 - 2E) J_4 = 0$$

$$\Rightarrow e_5 J_4 = 0$$

$$\therefore e_5 = E^3 - 2E = E(E^2 - 2)$$

$$= -E(E^2 + 1)$$

$$= -E e_4 - e_3$$

n=6:-

$$J_1 = 0, J_1 + (\lambda^2 h^2 - 2) J_2 + J_3 = 0$$

$$J_2 + (\lambda^2 h^2 - 2) J_3 + J_4 = 0$$

$$J_3 + (\lambda^2 h^2 - 2) J_4 + J_5 = 0$$

$$J_4 + (\lambda^2 h^2 - 2) J_5 + J_6 = 0$$

$$J_6 = 0$$

solving:-

$$-(\lambda^2 h^2 - 2)^4 J_5 + 3(\lambda^2 h^2 - 2)^2 J_5 - J_5 = 0$$

$$\Rightarrow (-E^4 + 3E^2 - 1) J_5 = 0$$

$$\Rightarrow e_6 J_5 = 0$$

$$\begin{aligned}
 e_6 &= -E^4 + 3E^2 - 1 \\
 &= E(E^3 + 2E) + E^2 - 1 \\
 &= -Ee_5 - e_4.
 \end{aligned}$$

$n=7:-$

$$\begin{aligned}
 J_1 &= 0, \quad J_1 + (\lambda^2 h^2 - 2)J_2 + J_3 = 0 \\
 J_2 + (\lambda^2 h^2 - 2)J_3 + J_4 &= 0 \\
 J_3 + (\lambda^2 h^2 - 2)J_4 + J_5 &= 0 \\
 J_4 + (\lambda^2 h^2 - 2)J_5 + J_6 &= 0 \\
 J_5 + (\lambda^2 h^2 - 2)J_6 + J_7 &= 0 \\
 J_7 &= 0
 \end{aligned}$$

solving:-

$$[(\lambda^2 h^2 - 2)^5 - 4(\lambda^2 h^2 - 2)^3 + 3(\lambda^2 h^2 - 2)]J_6 = 0$$

$$\Rightarrow (E^5 - 4E^3 + 3E)J_6 = 0$$

$$\begin{aligned}
 \Rightarrow e_7 J_6 &= 0 \quad \therefore e_7 = E^5 - 4E^3 + 3E \\
 &= E(E^4 - 3E^2 + 1) - E^3 + 2E \\
 &= -Ee_6 - e_5.
 \end{aligned}$$